

Get To The Body Shop

Agent001 Takes You Cabinet And Power Supply Shopping

If the combo of motherboard and CPU were said to be the sole of your PC then the cabinet and power supply (or PSU as it is called) duo would be the body. These are very important components but sadly ignored mostly due to ignorance on the part of both buyers and some vendors. Most people in India shop for a cabinet and a power supply the same way people buy potatoes; at least with potatoes some selection procedures exist; while for cabinets and PSUs they insist on the cheapest options available and a skimpy budget is allocated for the same. At the very most the consumer may ask the shopkeeper what the wattage rating of the PSU is, or how many fans does the cabinet come with. Vendors are, of course more concerned with profit margins and commissions so they don't always have your best interests in mind. However, as someone buying a PC, you should pay very close attention to both these components; because your shiny new processor, motherboard and graphics card will surely care. Inadequate power, dirty power, insufficient cooling, overheating and system stability are just some of the thoughts you should keep in mind.

As a technology enthusiast, I've seen, used and owned a number of PSUs, (or SMPS' as they're called), and cabinets. I've also seen el cheapo power supplies going *poof* and witnessed motherboard, CPU, RAM and HDD failure due to dirty power being supplied. The ill effects of overheating of CPUs, graphics cards, HDDs and even RAM as a result of substandard designs or product quality. The bottom line is in a country like ours bad electricity supply and overheating accounts for most of the PC failures which occur; the other major reason being mishandling.

Heating as you would know can be caused due to

two reasons; one, a particular component isn't getting adequately cooled, or two, improper air circulation which causes heat build up. In case of the former you had better watch out. The component that is overheating could get damaged and possibly damage other components as well. In the latter case, chances are you may not have problems for some time since no single component is overheating. But accumulated heat over time causes long term harm of reduced component life and could cause instability in general. No wonder server farms and mainframes run in low temperature air conditioned environments. However, even having an air-conditioner at home will not guarantee thermal bliss although it helps somewhat. The best way to eliminate heating is to use a well ventilated cabinet.

On to the role of the SMPS aka power supply. Not only does a power supply produce more heat when it cannot adequately power your PC's components, but it also reduces its life and lays the foundation for other problems. Suppose you have a 400 watt power supply and your PC

draws a steady 360 watts. This means this SMPS is working at 90 per cent of its rated load at all times. Given that most 400 watt power supplies can actually supply only 330 watts or so and you are in serious danger. Either the power supply will overheat and blow up, (I kid you not – this happens), or some of your components aren't getting the necessary power they need, which is akin to them being forced to do the same job with less energy. When I suggest a 650 watt power supply to a reader for a PC that I know will not consume more than 350 watts, I am taking into consideration the fact that stressing an SMPS with close to its maximum rated load is bad; also the higher the wattage the better the chance of the manufacturer using better materials to build it, although there's no hard and fast rule. Most manufacturers overstate wattage ratings, so two power supplies with rated outputs of 600 watts may well differ in terms of overall performance, heat generation, load capacity and even longevity.

When shopping for a cabinet or a power supply, please rid yourself of the mentality that these components don't do anything for performance and investing cash on them is just a necessary evil. Think of these binaries as twin protectors and facilitators of your PC, and it's always a good idea to keep at least 10 to 15 per cent of your overall budget for a PC on a good power supply and a well ventilated cabinet.

Some vendors will try to fool you by showing off the number of fans or vents a cabinet has – don't be fooled. Having more fans isn't always better. Airflow is *numero uno* priority here and two fans blowing air across each other or facing each other and blowing in opposite directions could actually be worse than no fans at all.

